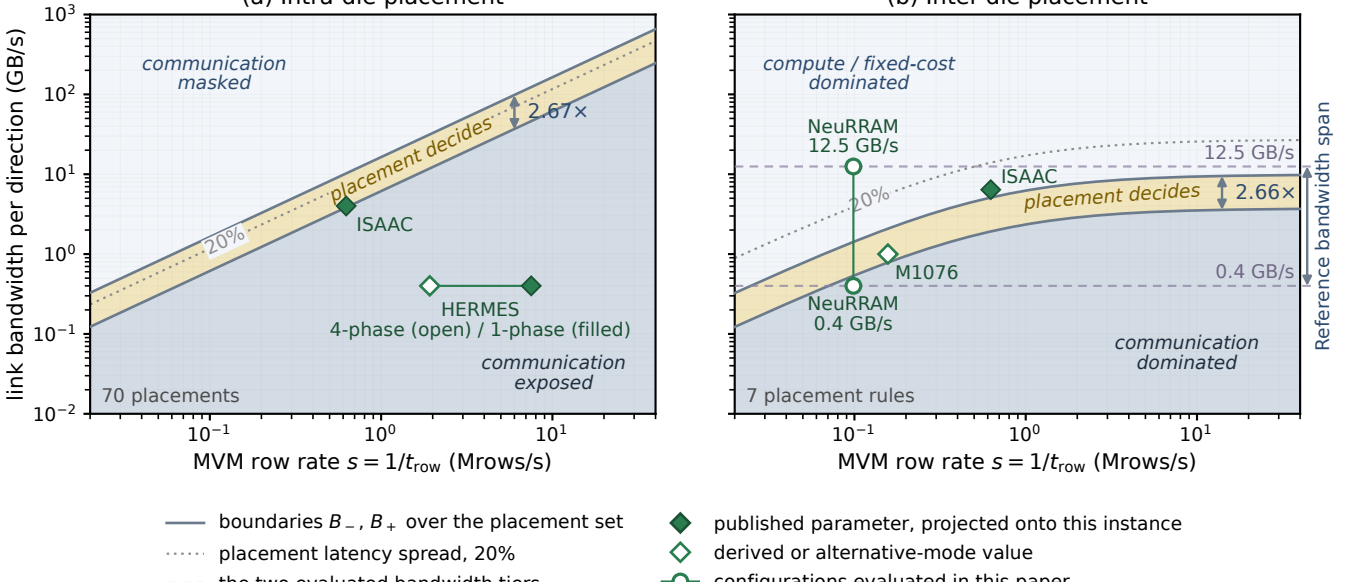
(a) Intra-die placement
(b) Inter-die placement
link bandwidth per direction (GB/s)
MVM row rate $s = 1/t_{row}$ (Mrows/s)
communication masked
placement decides
2.67×
20%
ISAAC
HERMES
4-phase (open) / 1-phase (filled)
communication exposed
70 placements
compute / fixed-cost dominated
NeuRRAM 12.5 GB/s
12.5 GB/s
2.66×
M1076
0.4 GB/s
NeuRRAM 0.4 GB/s
communication dominated
7 placement rules
Reference bandwidth span
boundaries B_-, B_+ over the placement set
placement latency spread, 20%
the two evaluated bandwidth tiers
published parameter, projected onto this instance
derived or alternative-mode value
configurations evaluated in this paper